USECASE FLOW DOCUMENTATION

UC-1: Login / Role-based Access

Actor(s)	Admin, Doctor, Receptionist, Patient
Goal	Allows a user to authenticate into the system and access a dashboard specific to their role.
Pre-condition	The user has a valid, active account registered in the system.
Main Flow	**1.** User navigates to the system login page. **2.** User enters username and password. **3.** User clicks the **Login** button. **4.** System validates the entered credentials against stored records. **5.** System identifies the user's role (Admin, Doctor, Receptionist, or Patient). **6.** System displays the role-specific dashboard with relevant menu options.
Alternate Flow	**Forgot Password** At Step 2, if the user does not remember their password: **2a.** User clicks the **Forgot Password** link on the login page. **2b.** User enters registered email/contact. **2c.** System sends a password reset link/OTP. **2d.** User resets the password and returns to **Step 1**. **Mandatory Fields Blank** At Step 3, if mandatory fields are left blank: **3a.** System displays a validation message. **3b.** System blocks submission and returns to **Step 2**. **Invalid Credentials** At Step 4, if authentication fails: **4a.** System displays an **Invalid credentials** error. **4b.** User may retry login (returns to **Step 2**) or cancel. **Account Locked** At Step 4, if the account is locked due to repeated failed attempts: **4a.** System displays a lockout message. **4b.** Login process ends until the account is unlocked.
Post-condition	User is successfully authenticated and directed to the appropriate role-based dashboard.

UC-2: Register Patient & Manage Medical Records

Actor(s)	Receptionist, Doctor, Admin
Goal	Allows hospital staff to register a new patient and maintain the patient's personal details, medical history, diagnosis, treatment, and prescriptions.
Pre-condition	The staff member is logged in with appropriate access rights.
Main Flow	**1.** Receptionist navigates to the **Patient Registration** screen. **2.** Receptionist enters the patient's personal details (name, age, gender, contact, address). **3.** Receptionist clicks **Save**. **4.** System validates the entered details and generates a unique Patient ID. **5.** Doctor opens the patient's profile during consultation. **6.** Doctor adds diagnosis, treatment notes, and prescriptions to the patient's medical record. **7.** System saves the updated medical record against the patient's profile.
Alternate Flow	**Update Existing Patient Details** At Step 1, if the staff wants to update an existing patient instead of registering a new one: **1a.** Staff searches for the patient by Patient ID or name. **1b.** Staff edits the required fields. **1c.** System saves the updated information, and the flow ends. **Mandatory Fields Missing** At Step 4, if mandatory fields (e.g., name) are missing: **4a.** System displays a validation error. **4b.** System blocks saving and returns to **Step 2**. **Invalid Contact Format** At Step 4, if the contact number format is invalid: **4a.** System prompts for correction. **4b.** Receptionist corrects the number and returns to **Step 3**. **Duplicate Patient Detected** At Step 4, if a potential duplicate patient (same contact number) is detected: **4a.** System warns the staff before saving. **4b.** Staff confirms or cancels, then returns to **Step 3**.
Post-condition	Patient is registered in the system with an accurate, up-to-date medical record accessible to authorized staff.

UC-3: Manage Doctor & Department Details

Actor(s)	Admin
Goal	Allows the Admin to manage doctor profiles, specializations, availability, and departments, and to assign doctors to departments.
Pre-condition	Admin is logged in with administrative access.
Main Flow	**1.** Admin navigates to **Doctor Management**. **2.** Admin adds a new doctor's name, specialization, and contact details. **3.** Admin navigates to **Department Management** and creates or selects a department. **4.** Admin assigns the doctor to the relevant department. **5.** Admin sets the doctor's availability (days and time slots). **6.** System saves the doctor and department information.
Alternate Flow	**Deactivate Doctor** At Step 1, if Admin wants to deactivate an existing doctor instead of adding a new one: **1a.** Admin selects a doctor profile. **1b.** Admin marks the doctor as inactive. **1c.** System removes the doctor from active appointment booking while retaining historical records, and the flow ends. **Mandatory Fields Missing** At Step 2, if mandatory fields such as specialization are missing: **2a.** System displays a validation error. **2b.** Admin corrects the details and returns to **Step 2**. **Department Deletion Blocked** At Step 3, if Admin attempts to delete a department that still has doctors assigned: **3a.** System blocks the action. **3b.** System displays a warning and returns to **Step 3**.
Post-condition	Doctor and department records are accurately maintained and reflected across the appointment booking module.

UC-4: Book, Reschedule, or Cancel Appointment

Actor(s)	Patient, Receptionist
Goal	Allows a patient or staff member to book a new appointment, reschedule an existing appointment, or cancel an appointment based on doctor availability.
Pre-condition	Patient is registered in the system and at least one doctor with available time slots exists.
Main Flow	**1.** User (Patient or Receptionist) navigates to **Book Appointment**. **2.** User selects the desired doctor, date, and an available time slot. **3.** User clicks **Confirm**. **4.** System checks slot availability. **5.** System books the appointment and updates the doctor's schedule. **6.** System sends a confirmation and schedules an appointment reminder.
Alternate Flow	**Reschedule Appointment** At Step 1, if the user wants to reschedule an existing appointment instead of booking a new one: **1a.** User opens an existing appointment. **1b.** User selects **Reschedule** and chooses a new available slot. **1c.** System updates the appointment and the associated reminder, and the flow ends. **Cancel Appointment** At Step 1, if the user wants to cancel an existing appointment: **1a.** User opens an existing appointment. **1b.** User clicks **Cancel** and confirms. **1c.** System updates the appointment status to **Cancelled** and frees the slot, and the flow ends. **Slot Already Booked** At Step 4, if the selected slot is already booked: **4a.** System displays **Slot not available**. **4b.** System prompts for another selection and returns to **Step 2**. **Past Date Selected** At Step 4, if the selected date is in the past: **4a.** System blocks booking with a validation message. **4b.** User returns to **Step 2** to select a valid date. **Concurrent Booking Conflict** At Step 5, if two users attempt to book the same slot simultaneously: **5a.** System allows only one booking to succeed. **5b.** The other user is returned to **Step 2** to select a different slot.

Post-condition	The appointment is accurately booked, rescheduled, or cancelled, and all related modules (doctor schedule, notifications) are updated.

UC-5: Admit and Discharge Patient

Actor(s)	Receptionist, Doctor
Goal	Allows hospital staff to admit a patient, allocate a room/bed, record treatment during hospitalization, and process discharge.
Pre-condition	Patient is registered and at least one bed is available in the required ward.
Main Flow	**1.** Receptionist navigates to **Admission** and selects the patient. **2.** Receptionist selects an available ward and bed and clicks **Admit**. **3.** System marks the patient as admitted and allocates the bed. **4.** Doctor adds treatment and procedure details during the hospitalization period. **5.** Upon recovery, Receptionist or Doctor initiates **Discharge**. **6.** User enters the discharge summary and discharge date, then confirms. **7.** System updates the patient status to **Discharged**, frees the bed, and triggers final bill generation.
Alternate Flow	**Bed Already Occupied** At Step 2, if the selected bed is already occupied: **2a.** System displays an error. **2b.** System prevents admission and returns to **Step 2** to select another bed. **Discharge Summary Blank** At Step 6, if the discharge summary is left blank: **6a.** System blocks discharge until it is completed. **6b.** User completes the summary and returns to **Step 6**.
Post-condition	Patient admission and discharge are accurately recorded, the bed status is updated in real time, and billing reflects the hospitalization charges.

UC-6: Generate Bill and Record Payment

Actor(s)	Receptionist, Patient
Goal	Allows billing staff to generate a consolidated bill for consultations, tests, treatments, and medicines, and to record corresponding payments.
Pre-condition	The patient has availed one or more billable hospital services.
Main Flow	**1.** Receptionist navigates to **Billing** and selects the patient. **2.** Receptionist adds all applicable charges (consultation, tests, treatment, medicines, room charges). **3.** Receptionist clicks **Generate Bill**. **4.** System calculates the total amount and creates an itemized bill. **5.** Receptionist or Patient selects a payment mode and enters the payment amount. **6.** System records the payment and updates the bill status (Paid / Partially Paid).
Alternate Flow	**Invalid Payment Amount** At Step 5, if the payment amount entered is zero or negative: **5a.** System displays a validation error. **5b.** User re-enters a valid amount and returns to **Step 5**. **Partial Payment** At Step 6, if a partial payment is made: **6a.** System updates the balance due. **6b.** System keeps the bill status as **Partially Paid**, and the flow ends.
Post-condition	An accurate bill is generated and linked to the patient's billing history, with payment status correctly reflected.

UC-7: Send Notifications and Reminders

Actor(s)	System (automated), Patient, Receptionist
Goal	Automatically generates and sends notifications or reminders for appointments, scheduled tests, and pending payments.
Pre-condition	A relevant event (appointment, test, or unpaid bill) exists with an associated date.
Main Flow	**1.** System's scheduled reminder job runs periodically. **2.** System identifies upcoming appointments, scheduled tests, and payments nearing their due date. **3.** System generates the appropriate notification for each event. **4.** System delivers the notification to the relevant user's Notification Center.
Alternate Flow	**Duplicate Reminder Prevented** At Step 3, if a reminder has already been sent for an event: **3a.** System does not generate a duplicate. **3b.** Flow returns to **Step 2** to process the next event. **Appointment Cancelled** At Step 2, if the related appointment is cancelled: **2a.** System removes the associated pending reminder. **2b.** Flow returns to **Step 2** to process the next event.
Post-condition	Relevant users are notified of upcoming appointments, tests, or dues in a timely manner.

UC-8: Generate Hospital Reports

Actor(s)	Admin, Doctor
Goal	Allows authorized users to generate reports related to patients, doctors, appointments, medical records, admissions, laboratory tests, and payments.
Pre-condition	User has report-generation access, and underlying data exists for the selected period.
Main Flow	**1.** User navigates to the **Reports** module. **2.** User selects the report type (e.g., Patient Report, Doctor Report, Payment Report). **3.** User specifies filters such as date range, doctor, or status. **4.** User clicks **Generate**. **5.** System compiles data from the relevant modules and displays the report. **6.** User optionally exports/downloads the report.
Alternate Flow	**No Data Found** At Step 5, if no data matches the selected filters: **5a.** System displays a **No records found** message. **5b.** User returns to **Step 3** to adjust the filters. **Large Date Range Handled** At Step 5, if the selected date range is very large: **5a.** System continues processing without failing. **5b.** System generates the report successfully within an acceptable time, and the flow continues to **Step 6**.
Post-condition	An accurate report is generated and made available for review or download.